

Sri Lanka Institute of Information Technology

Data warehousing and Business

Intelligence

Assignment 2



MAYADUNNA S M M T S – IT23584068

Year 03 Semester 01 - WE 01.01

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1.Introduction

This report presents the complete Business Intelligence solution developed for Assignment 2 of the IT3021 Data Warehousing and Business Intelligence module. The work undertaken in this assignment extends the PharmacyDW data warehouse, which was designed, implemented, and populated with data during Assignment 1, into a fully functional analytical platform. The solution encompasses the construction of a SQL Server Analysis Services (SSAS) multidimensional cube, the demonstration of fundamental OLAP operations through Microsoft Excel pivot tables, and the creation of four distinct Power BI reports that were subsequently published to the Power BI Service.

2. Data source for Assignment 2

The primary data source for this assignment is the PharmacyDW database, which was developed in the previous assignment. The database is structured using a star schema, where a central fact table is connected with several dimension tables. This structure is useful for performing data analysis from different perspectives and for generating reports in an efficient manner. In addition, the data is organised in a simplified form, which makes the analysis process more easier. The following sections will further explain the data integration process and the overall design of the data warehouse.

2.1 Source systems and staging layer

In this assignment, multiple data sources was used instead of a single file to simulate a real-world scenario. The data was divided into four sources including a CSV file for transactions, a SQL Server table for branch data, an Excel file for medicine and supplier details, and a text file for payment methods. These sources contain different types of data such as sales records, location details, and reference information. All the data was extracted using SSIS and loaded into staging tables first. In the staging area, basic cleaning and transformation was applied to make the data usable. After that, the processed data is loaded into the PharmacyDW warehouse, where dimension tables are filled before the fact table to maintain data consistency.

2.1.1 Source system

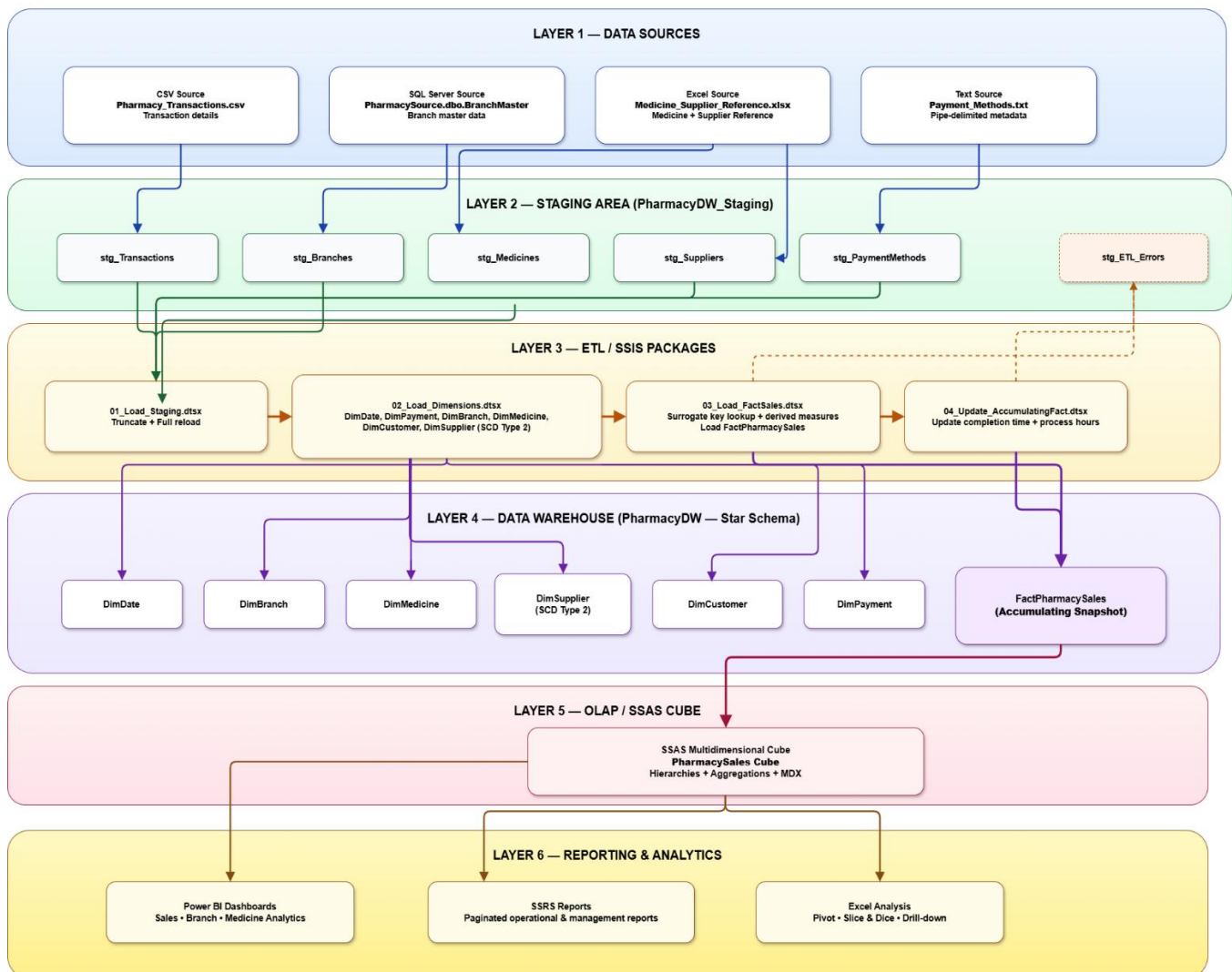
Source	File / Object	Format	Content	Staging Table
Source 1	Source1_Transactions.csv	CSV (Flat File)	50,000 raw sales transaction records with financial measures and customer demographics	stg_Transactions

Source 2	BranchMaster table in PharmacySource DB	SQL Server Database (OLE DB)	7 pharmacy branch records including geographic data and opening dates	stg_Branches
Source 3 – Sheet 1	Medicine_Supplier_Reference.xlsx	Excel (.xlsx)	148 medicine reference records with categories, dosage forms, and strength details	stg_Medicines
Source 3 – Sheet 2	Medicine_Supplier_Reference.xlsx	Excel (.xlsx)	8 supplier records with contact details, city, established year, and active status	stg_Suppliers
Source 4	Source4_Payment_Methods.txt	Pipedelimited Text File	3 payment method records with descriptions, digital flag, and cashback eligibility	stg_PaymentMethods

2.1.2 High-Level Architecture Diagram

The DW & BI solution follows a classic Kimball-style layered pipeline architecture consisting of six functional layers. Data flows from heterogeneous operational sources through ETL processing into a dimensional data warehouse, and from there into analytical and reporting tools

Sri Lanka Pharmacy Data Warehouse Modern Solution Architecture



2.2 Data warehouse design

The PharmacyDW warehouse was implemented as a star schema. The central fact table is FactPharmacySales, which records individual pharmacy sales transactions. This fact table is linked to six surrounding dimension tables: DimDate, DimBranch, DimMedicine, DimSupplier, DimCustomer, and

DimPayment. All analytical queries and aggregations are performed through the fact table by joining to the relevant dimensions using surrogate key relationships.

2.2.1 Dimension and Fact Table Summary

Table	Type	Primary Key	Natural Key	Description
DimDate	Dimension	DateKey (INT)	FullDate	Calendar dimension generated for Jan 2022 – Jun 2023. Supports time-based hierarchies.
DimBranch	Dimension	BranchKey	BranchID	Geographic branch dimension with Province -> District -> Branch hierarchy.
DimMedicine	Dimension	MedicineKey	MedicineID	Product dimension with Medicine Category -> Medicine Name hierarchy.
DimSupplier	SCD Type 2 Dimension	SupplierKey	SupplierName	Supplier dimension tracking history for email and city changes over time.
DimCustomer	Dimension	CustomerKey	Gender+AgeGroup+City	Derived demographic segment dimension. No CustomerID in source — segmented by gender, age group, and city.
DimPayment	Dimension	PaymentKey	PaymentMethod	Payment method reference dimension with digital and cashback flags.
FactPharmacySales	Accumulating Fact	SalesKey	TransactionID	Central fact table at transaction grain. Holds 6 FK references and all financial measures

2.2.2.Entity-Relationship Representation

The star schema ER diagram below illustrates the structural relationships among the seven tables. All foreign keys originate in FactPharmacySales and terminate at the corresponding dimension primary keys, establishing one-to-many cardinality in every case.

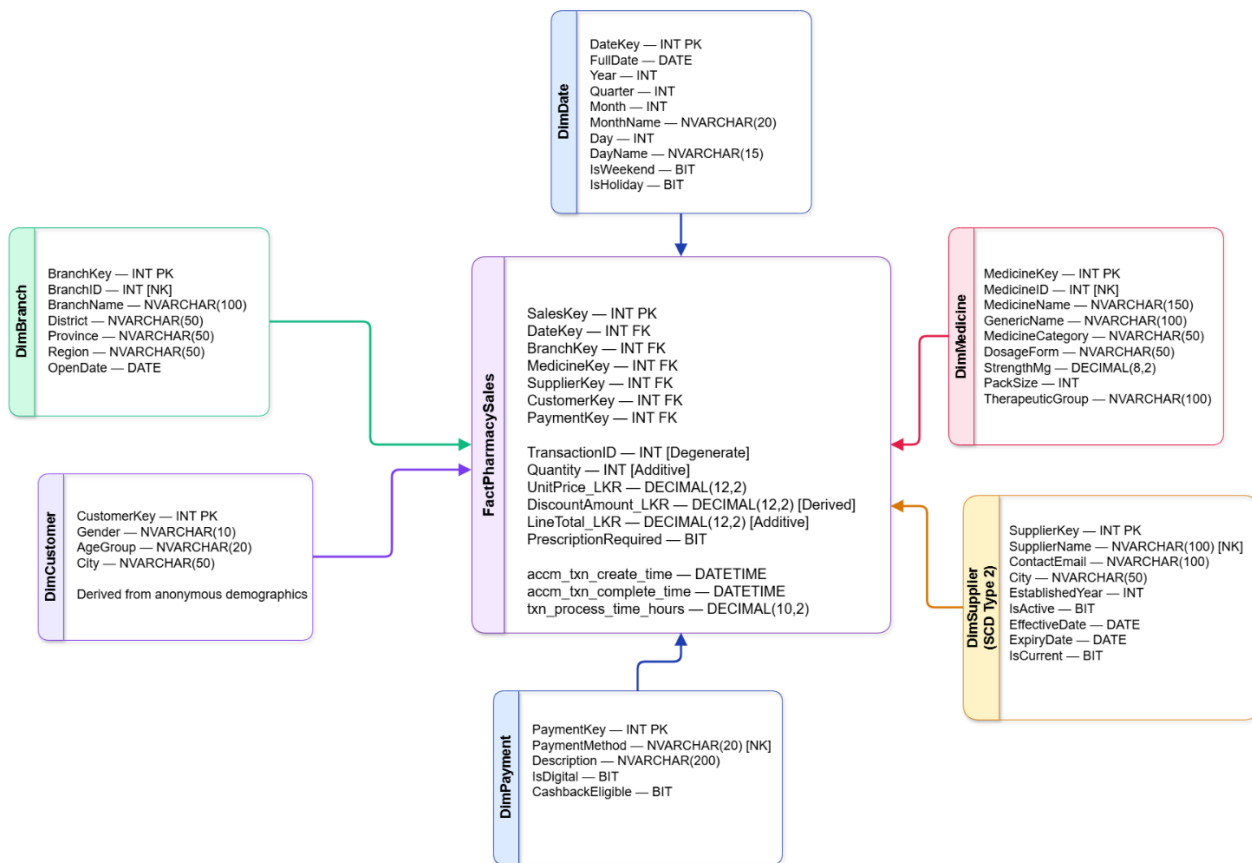
Sri Lanka Pharmacy Sales — ER Diagram



2.2.3 Entity Relationship Diagram

The warehouse is classical star schema topology wherein the FactPharmacySales table occupies the centre, surrounded by six conformed dimension tables. Foreign key relationships radiate outward from the fact table to each dimension, ensuring referential integrity and enabling join-free aggregation at query time.

PharmacyDW — Star Schema Dimensional Model



2.2.4 Fact Table Measures

Six numeric columns within FactPharmacySales carry the quantitative payload of the warehouse. Each column was assigned an appropriate default aggregation function during cube and report design.

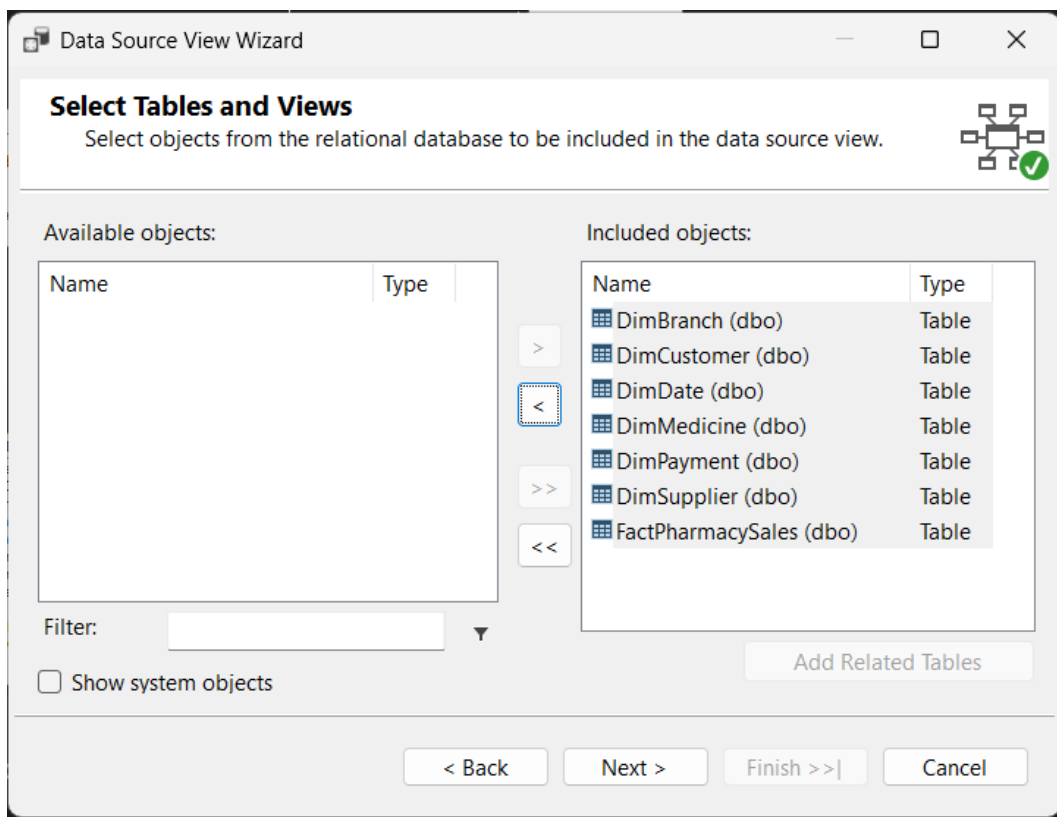
Measure Column	SQL Data Type	Semantic Meaning	Typical Aggregation
Quantity	INT	Number of medicine packs dispensed per transaction	SUM, AVG
UnitPrice_LKR	DECIMAL(12,2)	Selling price per pack in Sri Lankan Rupees	AVG, MIN, MAX
DiscountRate	DECIMAL(5,4)	Fractional discount applied to the line	AVG
DiscountAmount_LKR	DECIMAL(12,2)	Absolute discount value (UnitPrice × Qty × DiscountRate)	SUM
LineTotal_LKR	DECIMAL(12,2)	Net revenue after discount deduction	SUM, AVG
txn_process_time_hours	DECIMAL(10,2)	Elapsed hours between transaction creation and fulfilment	AVG

2.2.5 Hierarchies Available for SSAS

The warehouse is designed with different hierarchies to help analysis in SSAS cube. The Geography hierarchy in DimBranch has Province, District, and BranchName levels for drilling from region to branch. The Time hierarchy in DimDate has Year, Quarter, Month, and Day levels for analysing data over time. The Product and Customer hierarchies allow looking at sales by medicine category or customer age and gender, but in simple two levels only.

Hierarchy Name	Dimension	Levels	Business Purpose
Geography Hierarchy	DimBranch	Province -> District -> BranchName	Analyse sales by geographic region and drill into specific branches
Time Hierarchy	DimDate	Year -> Quarter -> Month -> Day	Analyse sales trends over time at different granularities
Product Hierarchy	DimMedicine	MedicineCategory -> MedicineName	Analyse sales by therapeutic category and drill into specific medicines
Customer Segment Hierarchy	DimCustomer	AgeGroup -> Gender	Analyse sales by customer demographic segment

Screenshot evidence for Step 1



3. SSAS cube implementation

3.1 Project configuration and deployment targets

The Analysis Services project was implemented as a Multidimensional and Data Mining Project in Visual Studio, designated as PharmacySales_Cube and configured to source data from the PharmacyDW relational database on the local SQL Server THEEKSHANA\IT23584068. Upon deployment to the local Analysis Services server, the SSAS database was assigned a unique identifier corresponding to the student's credentials. The following table summarises the key configuration elements of the SSAS implementation.

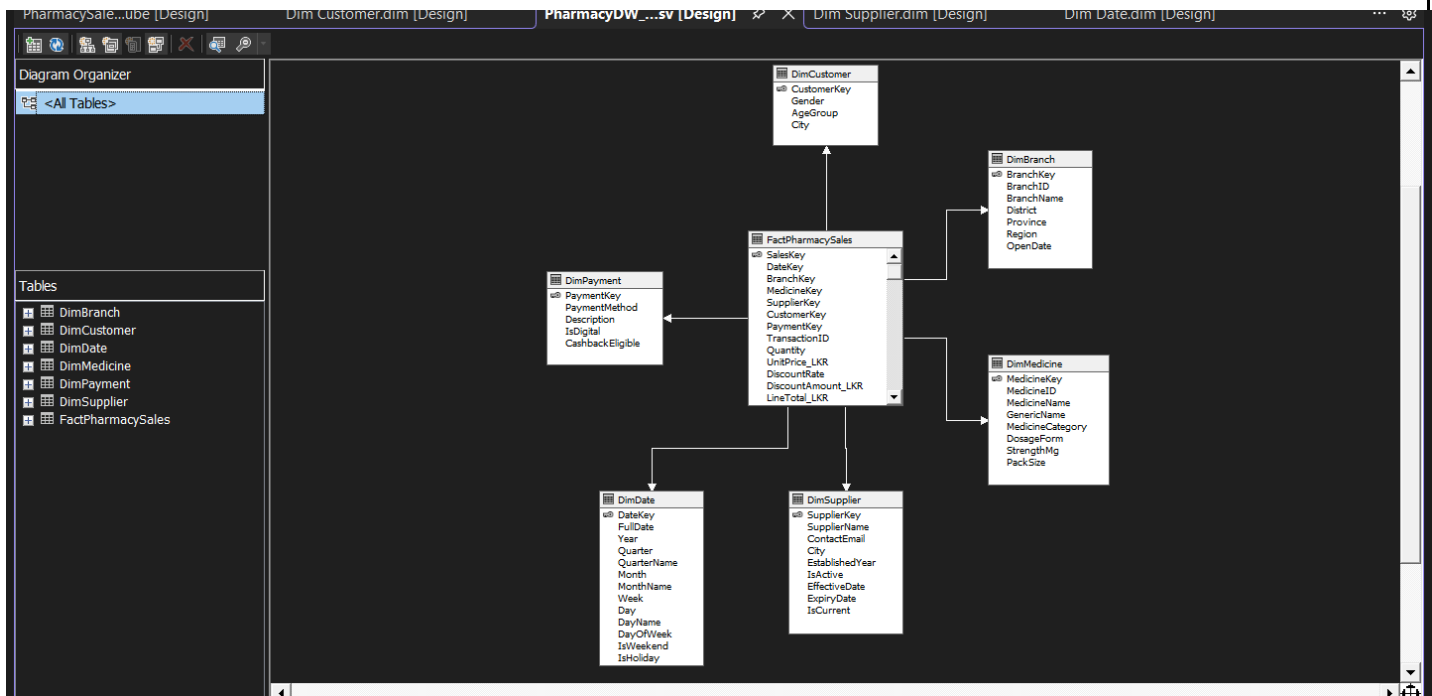
SSAS implementation configuration

Project element	Value used
Visual Studio project	PharmacySales_Cube_it23584068
SSAS cube name	PharmacySales_Cube
Data source name	Pharmacy DW
Data source view	PharmacyDW_DSV

Relational SQL source	localhost\THEEKSHANA\IT23584068
SSAS server	THEEKSHANA\SQLEXPRESS01
SSAS database	PharmacyDW_SSAS_IT23584068

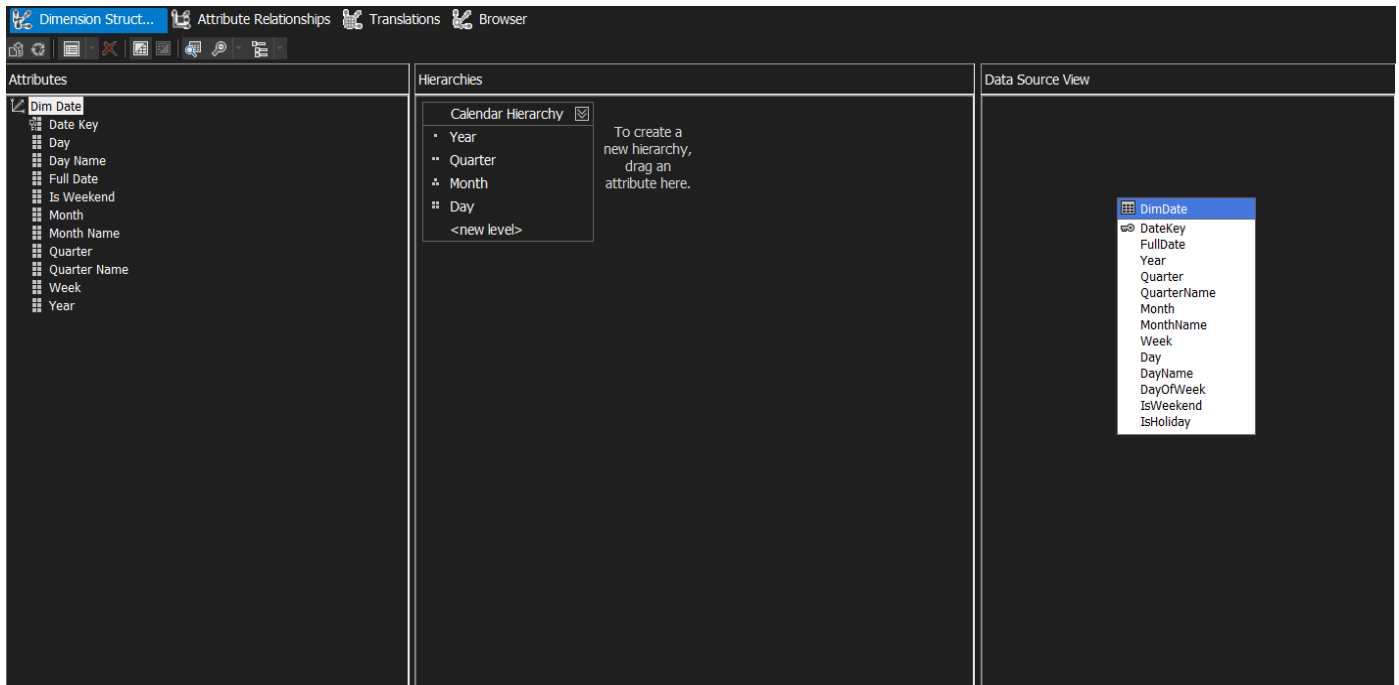
3.2 Data source, DSV, dimensions and hierarchies

A relational data source was established to connect the SSAS project with the PharmacyDW SQL Server database, using the Microsoft OLE DB Driver for SQL Server with Windows Authentication. The Data Source View (DSV) was then created including all seven warehouse tables (FactPharmacySales, DimDate, DimBranch, DimMedicine, DimSupplier, DimCustomer, and DimPayment). The DSV designer automatically detected most of the foreign key relationships between the fact table and dimension tables, however some relationships needed to be created manually by mapping foreign keys to their corresponding primary keys.



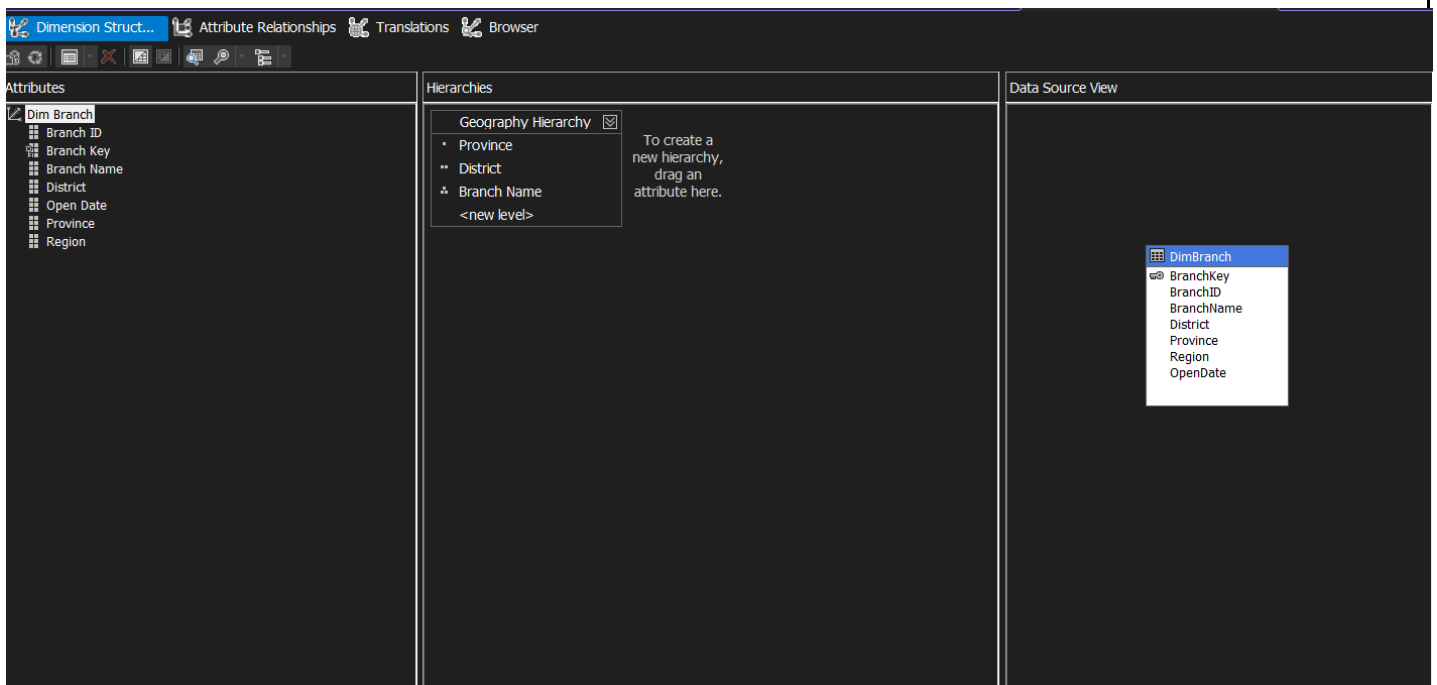
3.2.1 Calendar Hierarchy

Six dimensions were then created, one for each dimension table. The DimDate dimension was configured with DateKey as the key column and included attributes such as FullDate, Year, Quarter, QuarterName, Month, MonthName, Week, Day, DayName, and IsWeekend. Within this dimension, a Calendar Hierarchy was defined comprising four levels arranged as Year -> Quarter -> Month -> Day. This hierarchy enables temporal drill-down analysis from annual totals through to daily detail..



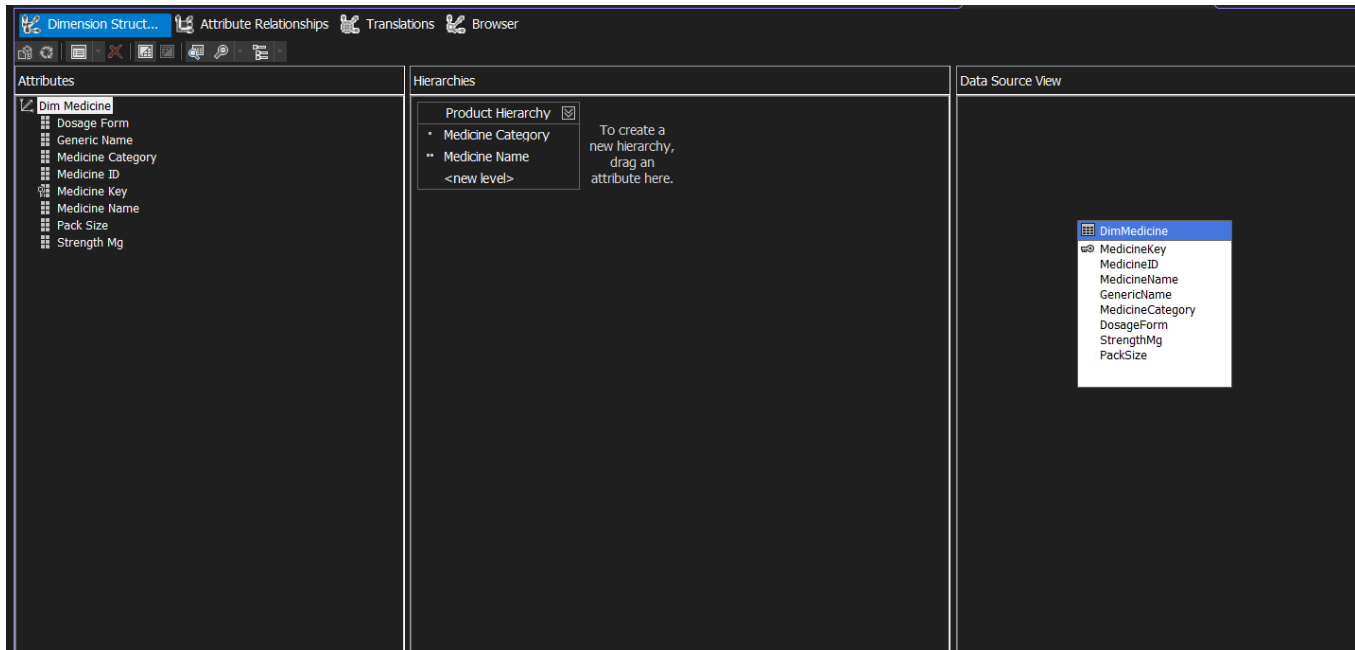
3.2.2 Geography Hierarchy

The DimBranch dimension was created with BranchKey as the key column, incorporating attributes including BranchID, BranchName, District, Province, Region, and OpenDate. A Geography Hierarchy was established within this dimension with three levels: Province at the top, followed by District, and BranchName at the lowest level. This hierarchy supports geographic analysis beginning at the provincial scope and narrowing to individual pharmacy branches.

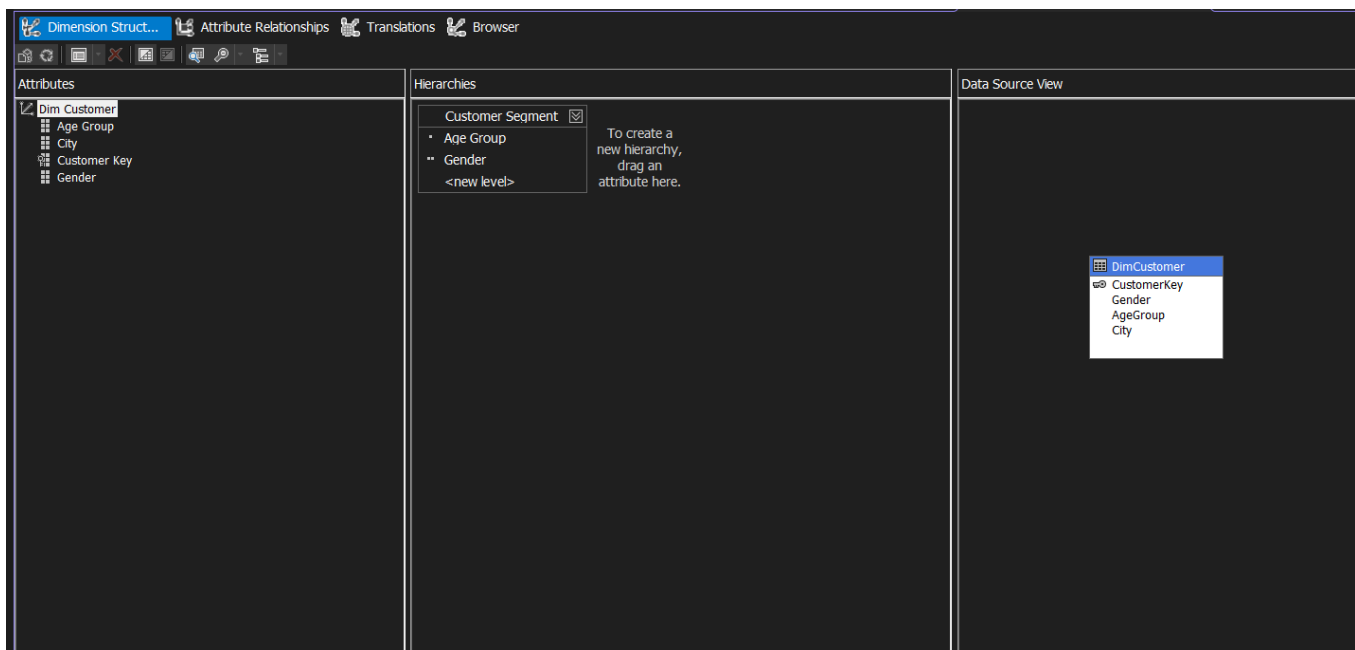


3.2.3 Product Hierarchy

The DimMedicine dimension was created with MedicineKey as the key column and included attributes such as MedicineID, MedicineName, GenericName, MedicineCategory, DosageForm, StrengthMg, and PackSize. A Product Hierarchy was defined with two levels: MedicineCategory -> MedicineName, enabling analysis from therapeutic category level down to specific medicines.



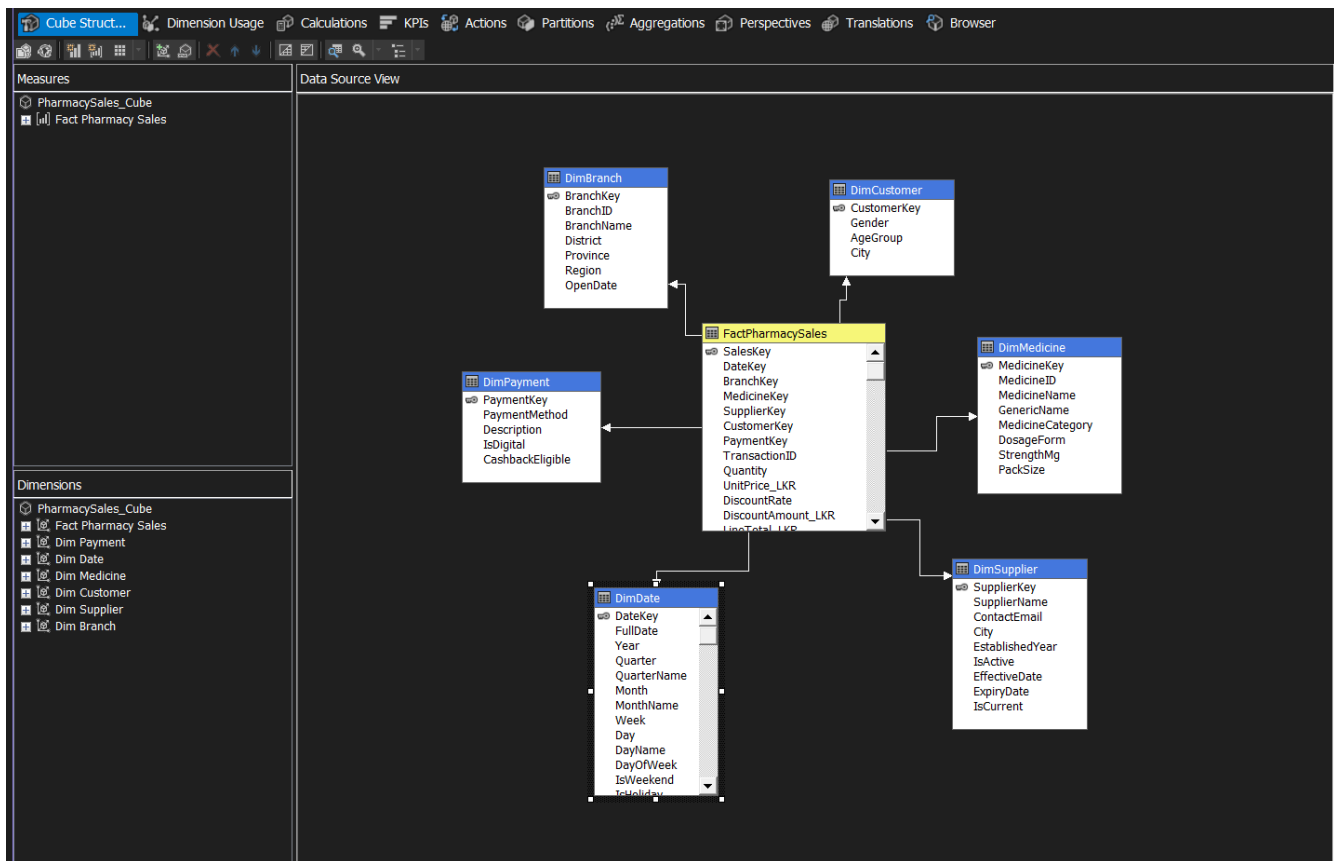
3.2.4 Customer Segment Hierarchy



3.3 Cube measures and dimension usage

The cube was created using the existing tables approach, with FactPharmacySales as the measure group table. Seven measures were configured with appropriate aggregation functions including Total Revenue (Sum of LineTotal_LKR), Total Quantity (Sum of Quantity), Average Unit Price (Average of UnitPrice_LKR), Total Discount Amount (Sum of DiscountAmount_LKR), Average Discount Rate (Average of DiscountRate), Transaction Count (Count of SalesKey), and Average Processing Hours (Average of txn_process_time_hours). Each measure was aligned to its analytical purpose for sales and operational analysis.

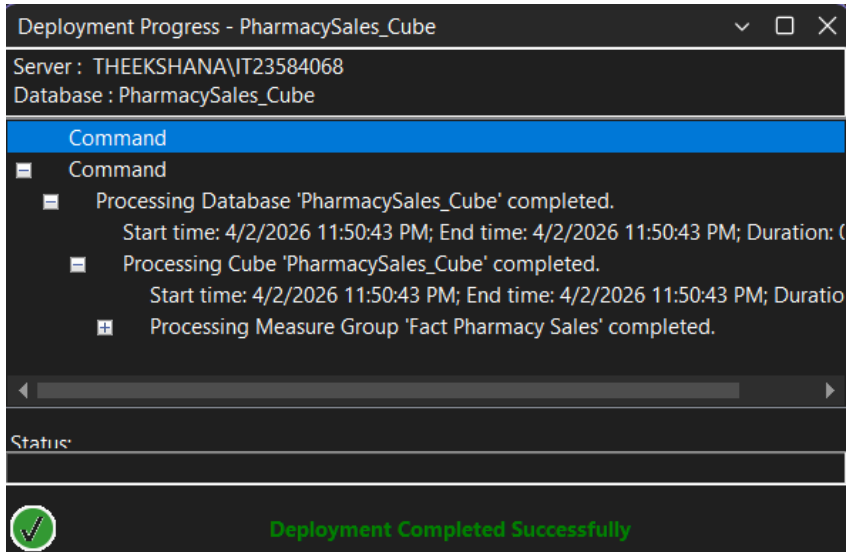
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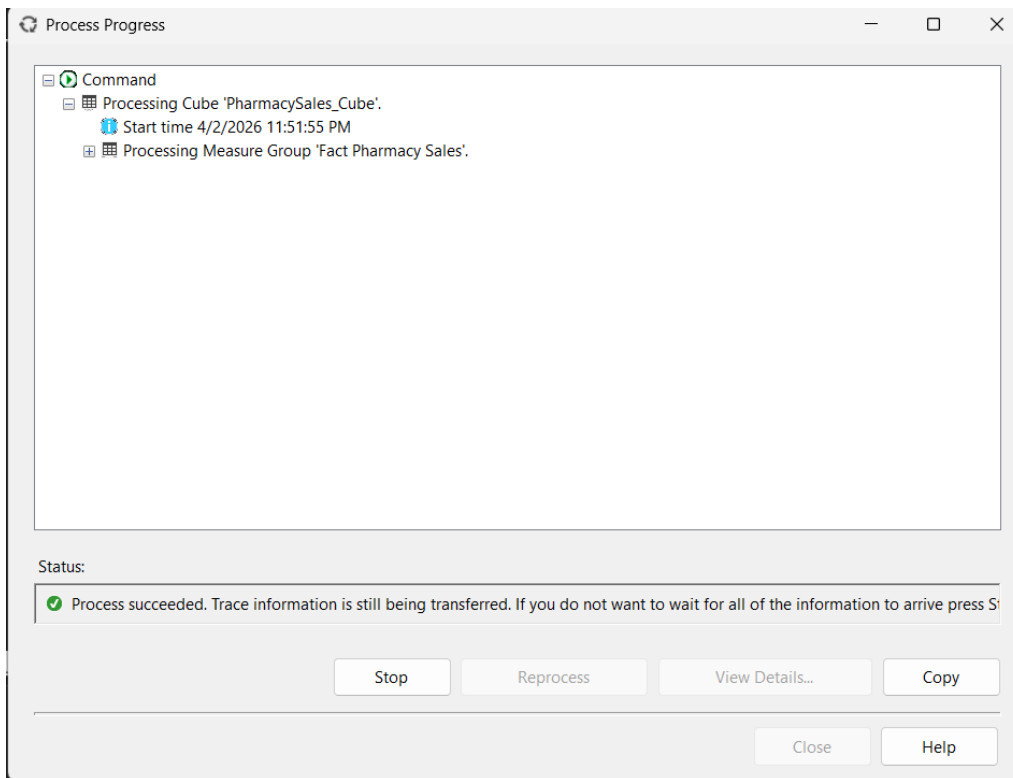
3.4 Deployment, processing and browser validation

The SSAS project was deployed to the local Analysis Services server instance through the project context menu, and the Deployment Progress window confirmed successful completion with zero errors and warnings. Following deployment, the cube was processed using Full Processing mode which loaded data from the PharmacyDW relational database into the multidimensional storage structures. The processing dialog confirmed that all six dimensions and the FactPharmacySales measure group were processed successfully with expected row counts reflected for each object.

Successful deployment of PharmcySales_Cube



Successful process progress of PharmcySales_Cube



SSAS Cube Browser showing granular sales data by Customer Key and Payment Method with Line Total LKR.

The screenshot shows the SSAS Cube Browser interface. The left pane displays the metadata tree for the 'PharmacySales_Cube' model, with 'Measures' expanded. The main pane shows a table with columns: Customer Key, Payment Method, and Line Total LKR. The table contains 18 rows of data, with the first row highlighted.

Customer Key	Payment Method	Line Total LKR
1601	Card	83684.44
1601	Cash	105949.21
1601	Insurance	26054.36
1602	Card	83112.44
1602	Cash	73085.05
1602	Insurance	37600.78
1603	Card	98648.67
1603	Cash	200669.75
1603	Insurance	83022.65
1604	Card	115034.45
1604	Cash	121240.88
1604	Insurance	5747.17
1605	Card	75158.4
1605	Cash	111482.46
1605	Insurance	45511.53
1606	Card	153466.17
1606	Cash	158871.48

SSAS Cube Browser showing sales count by Province, District, Branch Name, Medicine Category and Medicine Name.

The screenshot shows the SSAS Cube Browser interface. The left pane displays the metadata tree for the 'PharmacySales_Cube' model, with 'Measures' expanded. The main pane shows a table with columns: Province, District, Branch Name, Medicine Category, Medicine Name, and Fact Pharmacy Sales Count. The table contains 20 rows of data, with the first row highlighted.

Province	District	Branch Name	Medicine Category	Medicine Name	Fact Pharmacy Sales Count
Central	Kandy	Pharmacy - ...	Allergy	Calpol Cetirizin...	50
Central	Kandy	Pharmacy - ...	Allergy	Cardio Lorata...	58
Central	Kandy	Pharmacy - ...	Allergy	Cardio Lorata...	41
Central	Kandy	Pharmacy - ...	Allergy	Cardio Lorata...	49
Central	Kandy	Pharmacy - ...	Allergy	Glucocetirizin...	57
Central	Kandy	Pharmacy - ...	Allergy	Vita Loratadin...	45
Central	Kandy	Pharmacy - ...	Antibiotic	Amoxil Ciprofl...	43
Central	Kandy	Pharmacy - ...	Antibiotic	Amoxil Ciprofl...	54
Central	Kandy	Pharmacy - ...	Antibiotic	Calpol Levoflo...	54
Central	Kandy	Pharmacy - ...	Antibiotic	Cardio Ciprofl...	57
Central	Kandy	Pharmacy - ...	Antibiotic	Cardio Co-tri...	51
Central	Kandy	Pharmacy - ...	Antibiotic	Cardio Doxyc...	38
Central	Kandy	Pharmacy - ...	Antibiotic	Cardio Levoflo...	31
Central	Kandy	Pharmacy - ...	Antibiotic	Cardio Levoflo...	41
Central	Kandy	Pharmacy - ...	Antibiotic	Cipro Amoxicil...	47
Central	Kandy	Pharmacy - ...	Antibiotic	Cipro Amoxicil...	51
Central	Kandy	Pharmacy - ...	Antibiotic	Cipro Co-trim...	58

4. Demonstration of OLAP operations in Excel

4.1 Connecting Excel to the cube

Microsoft Excel was connected to the deployed SSAS cube to facilitate interactive OLAP analysis through pivot tables. The connection was established using the Data tab by selecting From Analysis Services option, then specifying the server name and selecting PharmacySales_Cube as the target database. Upon successful connection, a PivotTable report was created in a new worksheet with the PivotTable Fields panel displaying all available measures and dimensions, enabling Excel to issue MDX queries and retrieve pre-aggregated multidimensional data for interactive exploration.

Data Connection Wizard

Select Database and Table

Select the Database and Table/Cube which contains the data you want.

Select the database that contains the data you want:

PharmacySales_Cube

☒ Connect to a specific cube or table:

Name	Description	Modified	Created	Type
PharmacySales_Cube		4/2/2026 11:52:01 PM		CUBE

Cancel < Back Next > Finish

4.2 OLAP operation 1 - Roll-up

Roll-up aggregates sales data from the Month level up to Quarter level, and then to Year level. When viewing at Year level, all 18 months of transactions are aggregated into totals per year. 2022 Year Total Revenue = Sum of all monthly revenues across Jan 2022 – Dec 2022. 2023 Year Total Revenue = Sum of all monthly revenues across Jan 2023 – Jun 2023. This demonstrates how the Time hierarchy enables upward aggregation.

A4			
	A	B	
1	Row Labels	Line Total LKR	
2	2022		
3	1		
4	1	14327243.74	
5	2	13220670.52	
6	3	14261010.91	
7	2	42603319.79	
8	3	42472912.92	
9	4	43534016.38	
10	2023	89862823.61	
11	Grand Total	260281997.9	
12			
13			
14			

4.3 OLAP operation 2 - Drill-Down

Starting at Province level: 6 rows (Western, Central, Southern, North Western, Northern, Sabaragamuwa) Drilling into Western Province: reveals 2 districts (Colombo, Gampaha) Drilling into Colombo District: reveals branch Pharmacy - Colombo Fort This demonstrates the 3-level Geography Hierarchy providing progressive data exploration. Revenue and quantity data is correctly disaggregated at each level of the hierarchy..

	A	B	C
1			
2			
3			
4			
5	Row Labels	Line Total LKR	Quantity
6	⊕ Central	36026750.84	20843
7	⊕ North Western	37561479.29	21735
8	⊕ Northern	36672577.19	21421
9	⊕ Sabaragamuwa	37860964.66	21729
10	⊕ Southern	37075300.38	21310
11	⊖ Western		
12	⊖ Colombo		
13	Pharmacy - Colombo Fort	38041984.1	21754
14	⊕ Gampaha	37042941.41	21247
15	Grand Total	260281997.9	150039
16			
17			
18			

4.4 OLAP operation 3 – Slice

Slice operation fixes the Payment Method dimension to a single value. Selecting 'Insurance' slices the cube to show only insurance-paid pharmacy transactions. This reduces the 3-dimensional data (Time × Geography × Payment) to a 2D view. Total Revenue filtered to Insurance: shows which branches generate most insurance revenue. Useful for pharmacy management to analyses the value of insurance partnerships per branch.

	A	B	C
1	Payment Method	Insurance	
2			
3	Row Labels	Line Total LKR	Transaction ID
4	[-] Central		
5	[-] Kandy		
6	Pharmacy - Kandy City	5621155.78	81244
7	[+] North Western	5164650.89	76520
8	[+] Northern	5247392.41	82110
9	[+] Sabaragamuwa	5792586.08	80456
10	[+] Southern	5396814.46	77624
11	[+] Western	11231253.71	159511
12	Grand Total	38453853.33	557465
13			

4.5 OLAP operation 4 – Dice

Dice creates a sub-cube by applying multiple dimension constraints simultaneously. Applied constraints: Province = Western AND Year = 2022 AND PaymentMethod = Card. Result: Revenue per Medicine Category for card payments in Western province during 2022. This shows which therapeutic categories are most profitable in the Western province for card-paying customers. Compare this by changing Year to 2023 to see how category preferences changed year-over-year..

	A	B
1	Province	Western
2	Year	2022
3	Payment Method	Card
4		
5	Row Labels	Line Total LKR
6	Allergy	617007.13
7	Antibiotic	2255438.9
8	Antifungal	536120.62
9	Antiviral	818389.84
10	Cardiac	2790511.49
11	Cough & Cold	1603508.93
12	Dermatology	1740570.98
13	Diabetes	1038838.73
14	First Aid	1295222.44
15	Gastro	968438.08
16	Hormonal	929027.32
17	Painkiller	2170514.13
18	Respiratory	585554.05
19	Vitamin	2052259.81
20	Wellness	104743.75
21	Grand Total	19506146.2
22		
23		

4.6 OLAP operation 5 - Pivot

Pivot rotates the axes of analysis to change the perspective of the data. Before pivot: Rows = Province (6 rows), Columns = Medicine Category (15 columns) After pivot: Rows = Medicine Category (15 rows), Columns = Province (6 columns) The data values are the same but the orientation allows different comparisons: Before: Easy to compare categories within one province. After: Easy to compare provinces within one medicine category.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
4																	
5	Line Total LKR	Column Labels															
6	Row Labels	Allergy	Antibiotic	Antifungal	Antiviral	Cardiac	Cough & Cold	Dermatology	Diabetes	First Aid	Gastro	Hormonal	Painkiller	Respiratory	Vitamin	Wellness	Grand Total
7	Central	1248904.37	5042501.14	907171.89	1332317.46	4972531.42	3568982.86	2700625.3	2165002.53	2174718.89	1962366.67	1420476.71	3360950	854672.08	4068928.69	246600.83	36026750.8
8	North Western	1207566.06	4772727.71	797961.43	1596045.73	5605158.33	3490204.68	3199606.56	2121511.96	2524773.46	2213482.57	1346242.4	3483356.4	957184.87	4034555.4	211101.73	37561479.3
9	Northern	1300976.6	5047662.15	699188.32	1316155.8	4933482.87	3500343.79	2792832.02	2359822.02	2277110.07	2055491.72	1449956.23	3409135.06	1003744	4305305.31	221371.24	36672577.2
10	Sabaragamuwa	1293210.88	4752871.13	972893.19	1627517.17	5443839.25	3427737.71	3125869.64	2312553.86	2418013.79	1886280.35	1358034.23	3662860.4	1202687.9	4165112.66	211482.49	37860964.7
11	Southern	1265692.2	4589235.55	957744.02	1369750.51	5593210.89	3444345.4	2792726.22	2333996.5	2206112.11	2047415	1593312.2	3195123.02	1122626.5	4330343.39	233666.85	37075300.4
12	Western	2526535.83	9751400.28	1862944.8	3022950.76	10181443.1	6467058.2	5927975.68	4347875.3	5085114.61	4356411.83	3121504.2	7095507.27	2230241.7	8658551.81	449410.13	75084925.5
13	Grand Total	8842885.94	33956398	6197903.6	10264737.4	36729665.9	23898672.6	20539635.4	15640762.2	16685842.9	14521448.1	10289526	24206932.2	7371157.1	29562797.3	1573633.3	260281998
14																	
15																	
16																	
17																	

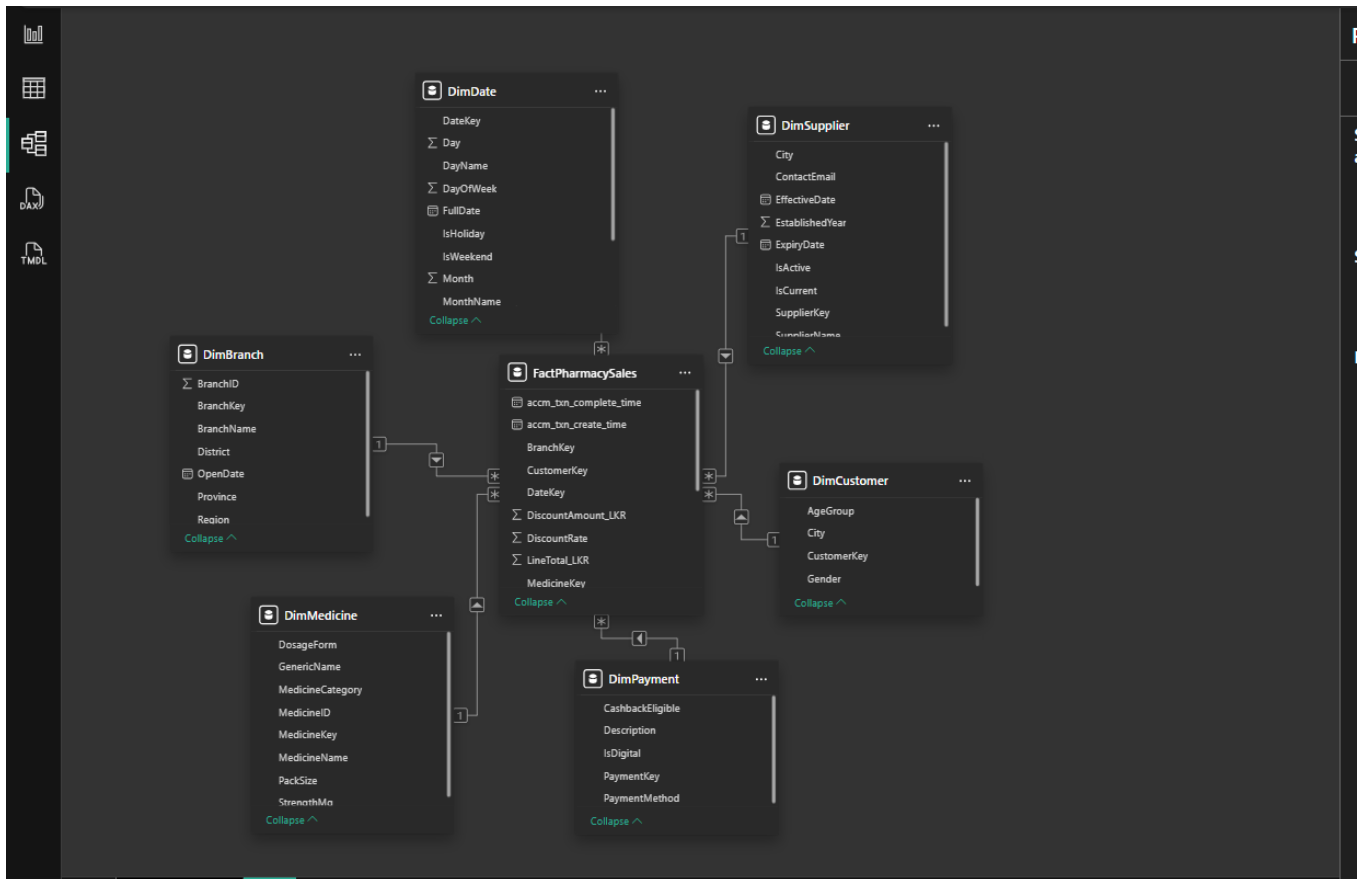
	A	B	C	D	E	F	G	H
4								
5	Line Total LKR	Column Labels						
6	Row Labels	Central	North Western	Northern	Sabaragamuwa	Southern	Western	Grand Total
7	Allergy	1248904.37	1207566.06	1300976.6	1293210.88	1265692.2	2526535.83	8842885.94
8	Antibiotic	5042501.14	4772727.71	5047662.15	4752871.13	4589235.55	9751400.28	33956397.96
9	Antifungal	907171.89	797961.43	699188.32	972893.19	957744.02	1862944.78	6197903.63
10	Antiviral	1332317.46	1596045.73	1316155.8	1627517.17	1369750.51	3022950.76	10264737.43
11	Cardiac	4972531.42	5605158.33	4933482.87	5443839.25	5593210.89	10181443.12	36729665.88
12	Cough & Cold	3568982.86	3490204.68	3500343.79	3427737.71	3444345.4	6467058.2	23898672.64
13	Dermatology	2700625.3	3199606.56	2792832.02	3125869.64	2792726.22	5927975.68	20539635.42
14	Diabetes	2165002.53	2121511.96	2359822.02	2312553.86	2333996.5	4347875.3	15640762.17
15	First Aid	2174718.89	2524773.46	2277110.07	2418013.79	2206112.11	5085114.61	16685842.93
16	Gastro	1962366.67	2213482.57	2055491.72	1886280.35	2047415	4356411.83	14521448.14
17	Hormonal	1420476.71	1346242.4	1449956.23	1358034.23	1593312.2	3121504.2	10289525.97
18	Painkiller	3360950	3483356.4	3409135.06	3662860.4	3195123.02	7095507.27	24206932.15
19	Respiratory	854672.08	957184.87	1003743.99	1202687.91	1122626.52	2230241.71	7371157.08
20	Vitamin	4068928.69	4034555.4	4305305.31	4165112.66	4330343.39	8658551.81	29562797.26
21	Wellness	246600.83	211101.73	221371.24	211482.49	233666.85	449410.13	1573633.27
22	Grand Total	36026750.84	37561479.29	36672577.19	37860964.66	37075300.38	75084925.51	260281997.9
23								
24								

5. Power BI Reports

5.1 Data preparation and model relationships

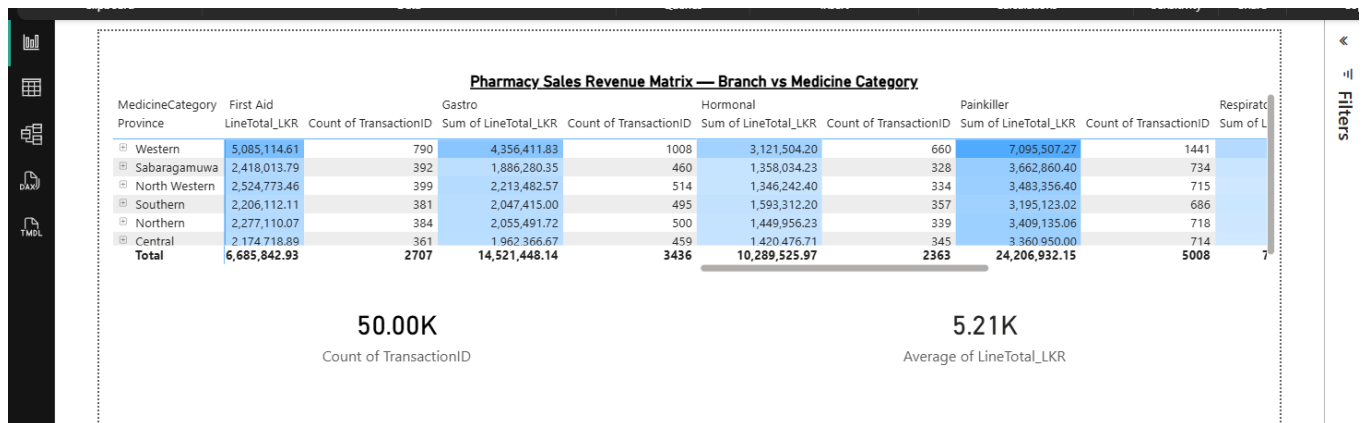
Power BI Desktop was used to connect directly to the PharmacyDW SQL Server database using the SQL Server data connector in Import mode. All seven tables (DimDate, DimBranch, DimMedicine, DimSupplier, DimCustomer, DimPayment, and FactPharmacySales) were selected and loaded into the Power BI data model. The Model view was examined to verify that the star schema relationships were correctly established, with each dimension table on the one-side of a many-to-one relationship with FactPharmacySales through appropriate surrogate key columns. Where relationships were not

automatically detected, they were manually created, resulting in a star pattern with FactPharmacySales at the centre and six dimension tables radiating outward.



5.2 Report 1 - Matrix visual

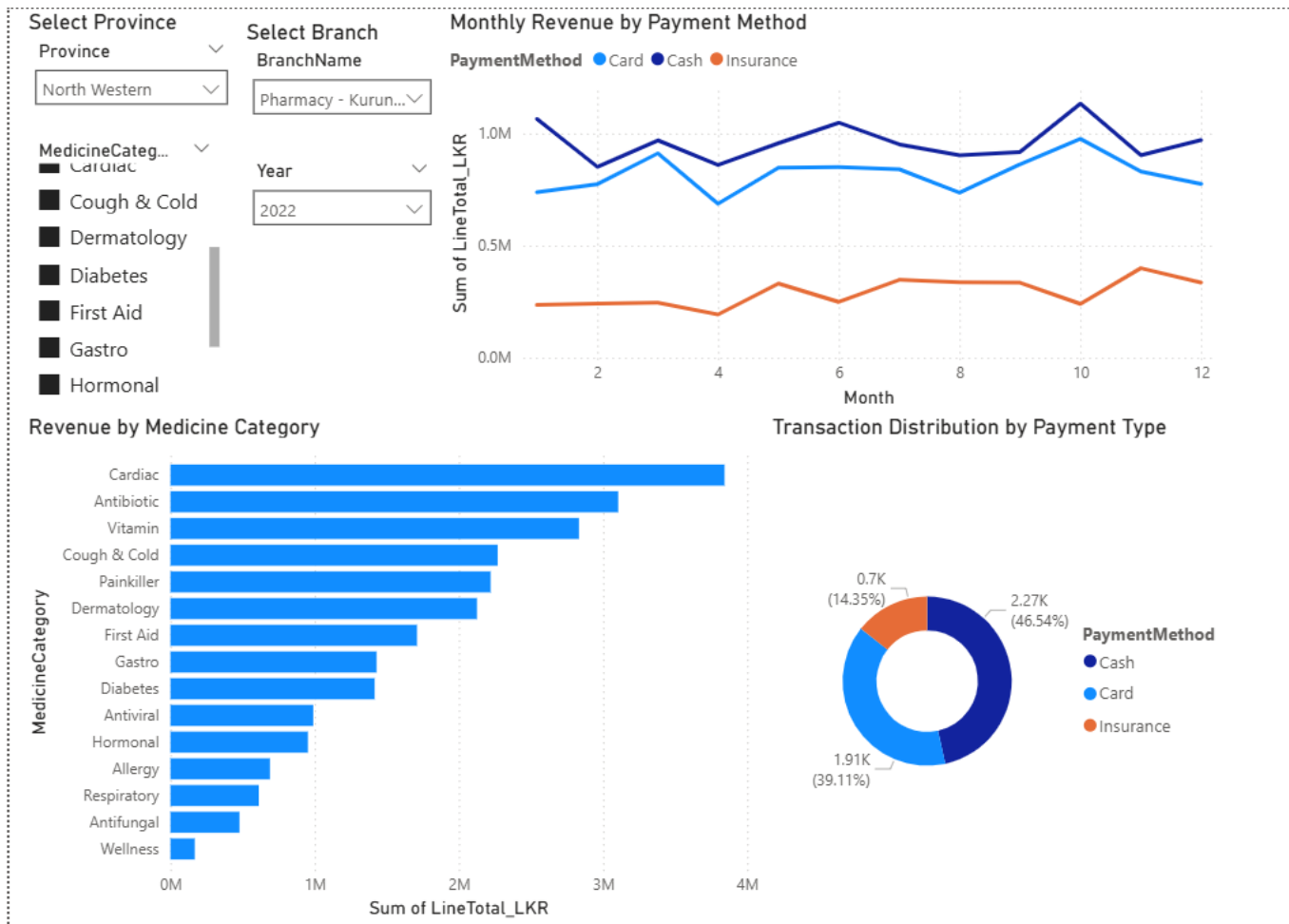
The first report page was designed using the Matrix visual to present a detailed tabular cross-tabulation of pharmacy sales data. The Matrix was configured with Province and BranchName in the Rows, MedicineCategory in the Columns, and Total Revenue, Total Quantity, and Transaction Count in the Values well, with row and column subtotals enabled and conditional formatting applied to revenue values using colour scale. Two Card visuals were positioned above the matrix to display overall Total Revenue and Transaction Count as headline summery figures, fulfilling the assignment requirement for a Matrix visual with distinctive pharmacy-specific cross-tabulation of geographic branches against therapeutic medicine categorie.



5.3 Report 2 - Multiple slicers with cascading filter

The second report page was constructed as an interactive, slicer-driven analytical dashboard with cascading filter behaviour. Four slicers were positioned on the left side including Province and BranchName (with automatic cascading as they belong to same DimBranch table), MedicineCategory in list style, and Year in dropdown style. The report canvas includes three analytical visuals that respond to slicer selections: a clustered bar chart displaying Total Revenue by MedicineCategory, a line chart plotting monthly Total Revenue against Month segmented by PaymentMethod, and a donut chart showing Transaction Count distribution across payment methods. This report satisfies the assignment requirement for multiple slicers with cascading filters and multiple visuals presenting data-driven insights. The page contains three visuals,

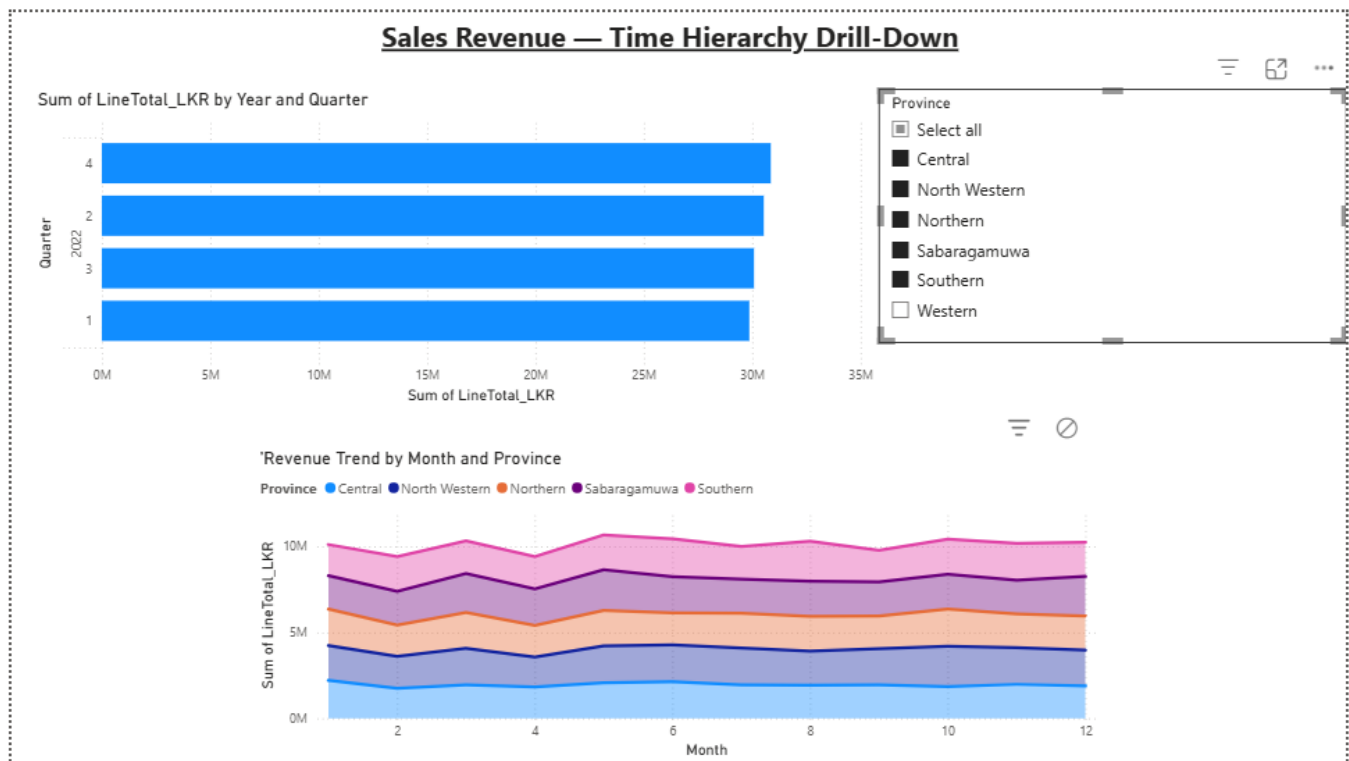
- Clustered bar chart was created using MedicineCategory from DimMedicine and Total Revenue (LKR) to identify top-performing therapeutic categories.
- Line chart was created using Month from DimDate and Total Revenue (LKR), with PaymentMethod from DimPayment as the legend, to show monthly revenue trends by payment type.
- Donut chart was created using PaymentMethod from DimPayment and Transaction Count to display the proportion of transactions across Card, Cash, and Insurance methods.



5.4 Report 3 – Drill-down report

The third report page was designed to enable hierarchical drill-down exploration of sales revenue across the Time Hierarchy. A clustered bar chart was created with Total Revenue on the X-axis and a hierarchical field structure on the Y-axis including Year, Quarter, and Month from DimDate in descending order of granularity. This hierarchical axis configuration activates the built-in drill-down capability in Power BI, allowing users to navigate from year-level revenue (2022 and 2023) down to quarterly figures and then to individual monthly values by clicking on specific bars.

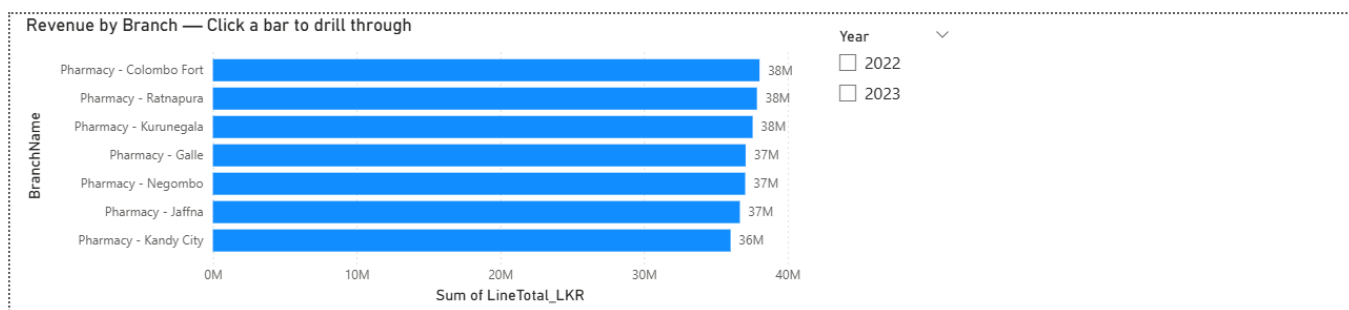
A second visual, a stacked area chart, was added to complement the drill-down bar chart by plotting monthly Total Revenue against Month with Province as the legend to show how revenue trends differ across geographic regions over time. A Province slicer was also included to filter both visuals simultaneously. This report demonstrates the ability to explore data hierarchically from year through quarter to month, fulfilling the corresponding assignment requirement.



5.5 Report 4 – Drill-through report

Summary page

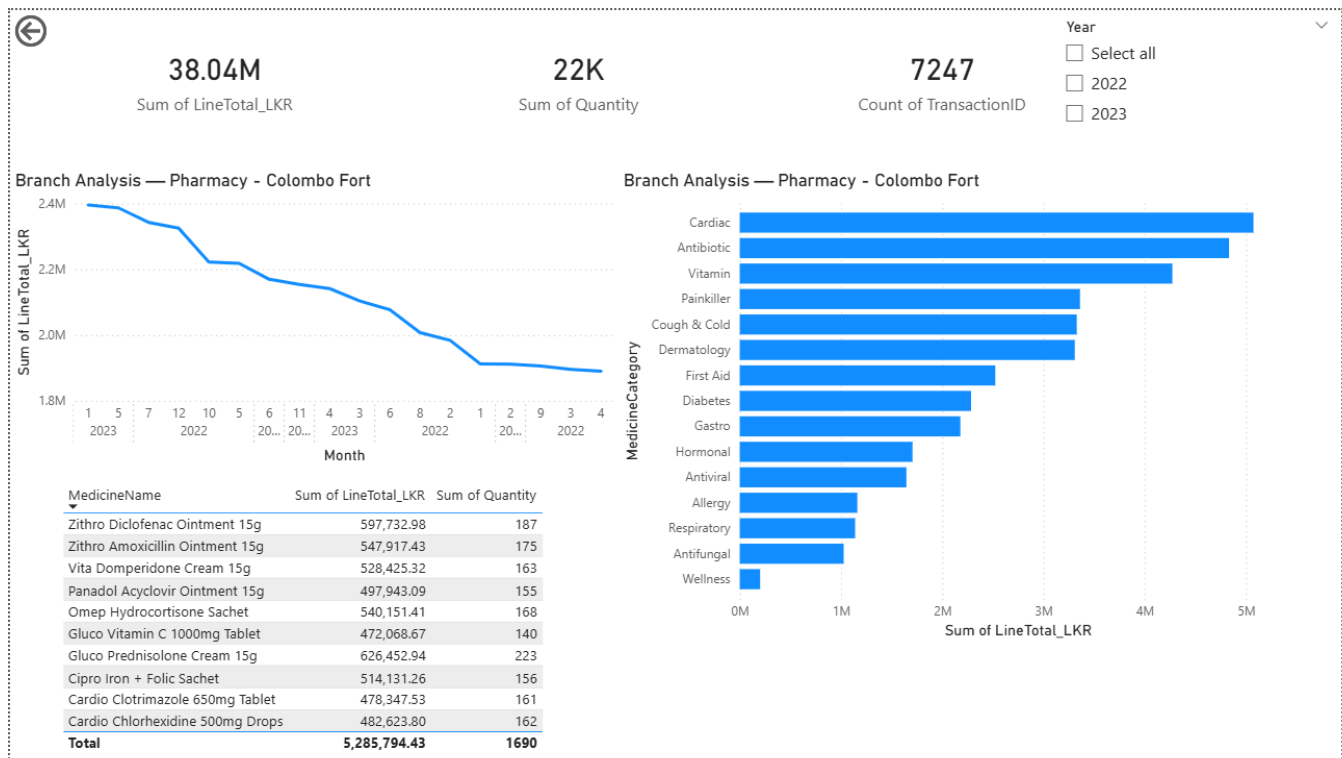
The summary page contains a bar chart that shows Total Revenue (LKR) by BranchName for all seven pharmacy branches. The chart is placed on the top of the page and colours each bar different. Users can right click on any branch bar and choose the drill through option to move to the Branch Detail page. This gives a quick overview of the branch performance.



Branch Detail page

In the Branch Detail page a drill-through filter on BranchName is set so that when a user goes from the summary page the page is automatically filtered to the selected branch. The page has three Card visuals that show the branch specific Total Revenue, Transaction Count and Total Quantity. A line chart displays

monthly revenue trend, a bar chart shows revenue by MedicineCategory, a table lists the top medicines with their quantities and a Yea slicer lets the user filter the view by time. A back button appears automatically in Power BI to return to the summary page.



5.6 Publishing to Power BI Service

After completing all four report pages, the report was published to the Power BI Service through the Publish button and accessed via app.powerbi.com. All interactive features including cascading slicers, drill-down, and drill-through functionality were tested and confirmed to be working correctly in the online environment.

Publishing to Power BI

✓ Success!

[Open 'POWERbi.pbix' in Power BI](#)

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Did you know?

You can create a portrait view of your report, tailored for mobile phones. On the **View** tab, select **Mobile Layout**. [Learn more](#)

Got it

